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Chapter 1

Blockchain Based Data Provenance for Trusted Artificial Intelligence

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Data reliability is a prerequisite for the development of effective and trusted Artificial Intelligence (AI) systems in industrial environments. Unfortunately, industrial data tend to be unreliable for a variety of reasons (e.g., environmental influence, background noise, and sensor failures). This chapter presents the advantages of blockchain technologies for tracking, tracing, and boosting the reliability of industrial data. It also reviews different blockchain solutions for digital manufacturing, including data provenance and reliability solutions. The chapter ends-up presenting a complete solution for tracing data and metadata of AI algorithms for industrial applications. The solution ensures the use of "sealed" AI algorithms leveraging the properties that render blockchains resilient to tampering. Its main function is to persist the metadata of the algorithms, as well as their outcomes (e.g., prediction

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and classification outcomes). As such, it facilitates the implementation strategies that secure AI systems in production lines and boost a trusted AI environment in manufacturing applications.

1.1 Introduction

During the past decade industrial organizations have accelerated their digital transformation and their transition to the fourth industrial revolution (Industry 4.0). This transition is empowered by the deployment of a proliferating number of internet connected systems (e.g., embedded sensors and other Internet of Things (IoT) devices) and of Cyber Physical Production Systems (CPPS) in the manufacturing shopfloor. IoT devices and CPPS systems enable the collection of digital data from physical production processes [1]. Likewise, the analysis of these data drives several optimizations in areas like logistics, quality management, assets maintenance, process control and supply chain management. These optimizations are implemented based on advanced analytics technologies, including BigData analytics, Machine Learning (ML) and Artificial Intelligence (AI). The outcomes of such analytics are used to optimize production processes and to improve manufacturing decisions.

In recent years, Artificial Intelligence (AI) is considered the most disruptive and impactful digital enabler of Industry 4.0. AI systems and algorithms are nowadays enabling manufacturing use cases with a proven Return on Investment (ROI), such as predictive maintenance [2], predictive quality management (Quality 4.0) [3], zero-defect manufacturing [4] and generative product design [5]. The rise of AI in manufacturing use cases is driven by the explosion of available data points, the accelerated increase of available computational capacity, as well as advances in AI software frameworks and tools. Data availability is a key prerequisite for the development of effective AI systems in manufacturing, which is the main reason why most national strategies for AI consider the transition to a data economy as a critical success factor for the AI era [6].

Unfortunately, industrial enterprises still struggle to collect well-structured and high-quality datasets. Industrial data tend to be fragmented across different "siloed" systems such as CPPS systems, business information systems (e.g., ERP (Enterprise Resource Planning) system), asset maintenance systems (e.g., Computerized Maintenance Management System (CMMS)), historian databases, automation systems (e.g., Supervisory Control and Data Acquisition (SCADA) systems) and more. These systems comprise structured (e.g., sensor data), semi-structured (e.g., operations records) and unstructured data (e.g., product images), while having different semantics, formats, and representations. Likewise, some of the data

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have high velocity (e.g., sensor data), while others are simply high-volume data at rest (e.g., transactional data). The above-listed factors make their integration and utilization in AI use cases very challenging. To alleviate these interoperability and integration challenges, several interoperability and BigData management technologies have been developed, including technologies based on industrial standards for data interoperability. Nevertheless, even with interoperability solutions at hand, industrial organizations must also confront one more challenge, namely the unreliability of industrial data.

Data reliability is one of the most important issues in industrial environments, especially when it comes to developing and deploying AI systems. Without reliable data, it is almost impossible to develop effective and high-performance AI algorithms due to the well-known GIGO (Garbage In Garbage Out) phenomenon [7]. Furthermore, the use of unreliable data for training AI algorithms is likely to lead to biased and unreliable systems. There are many reasons why data reliability is very challenging in industrial environments. Specifically, industrial data collection is an inherently unreliable process because of:

- Environmental influences like high or low temperatures, humidity, moisture, and air pressure factors.
- Background noise such as noise pollution or interference (e.g., alarms, extraneous speech) and electrical noise from devices like motors, cooling devices, air conditioning, and power supplies.
- Faulty or inaccurate sensors i.e., sensing systems with poor precision.
- Dying battery of a system that compromises its ability to operate properly and provide reliable measurements.
- Compromised or attacked devices that produce biased or fake data due to adversarial attacks (e.g., data modification, false information injection).
- Compromised AI or BigData analytics algorithms, such as algorithms under poisoning or evasion attacks [8].

To alleviate data unreliability, industrial enterprises take various measures, including data cleansing of databases, quality control on the various data sources, as well as reconciliation of conflicting information towards a single version of the truth. In the area of data security and resilience, it is important to ensure that data infrastructures are cyber-resilient and cannot be tampered. In this direction, the use of distributed ledger technologies (most notably blockchains) is suggested in several research works. Blockchain technology is widely known as the underlying technology of blockbuster crypto-currencies (e.g., Bitcoin, Ethereum) and other crypto-assets. Nevertheless, it is increasingly used in other applications beyond digital finance, including industrial applications in sectors like manufacturing, energy, agriculture, and supply chain management [9].

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Blockchain infrastructures provide the means for decentralized data management, including ways for decentralized data operations and transactions, such as CRUD (CReate Update Delete) operations. They come with many compelling properties for reliable data operations including:

- **Decentralized operation – No single point of failure:** Blockchain infrastructures operate in a highly distributed fashion and do not rely on a trusted third party for the validation of data transactions. This architectural property makes them much more difficult to be compromised, as they have no single point of failure. It also ensures their around-the-clock availability.
- **Tamper-resilience:** Distributed ledgers feature anti-tampering properties. Data written in a distributed ledger requires a next-to-impossible investment in resources to be changed. Depending on the algorithm employed to achieve consensus within a blockchain network, there are no good incentives to attempt to tamper the state of the blockchain. This is a foundation for data reliability, as blockchain data cannot be changed by adversarial parties.
- **Data transparency and auditability:** Transactions persisting (meta)data on a blockchain are transparent and accessible to all members (peers) of a blockchain network. As such they are auditable and open to scrutiny of other participants to the distributed ledger infrastructure.
- **Security:** Blockchain infrastructures offer very secure integrity protection mechanisms, including data hashing and cryptographical linking among the various blocks. This boosts their tamper-proof nature and minimizes security risks. Likewise, it is not possible to hack a blockchain by attacking few of its nodes. Blockchains support consensus mechanisms (e.g., majority voting), which require an absolute majority of nodes to agree on changes to the blockchain contents. In this way, they are resilient against cyber-attacks that could compromise one or more nodes.

These properties make blockchain infrastructures appropriate for the implementation of manufacturing use cases that involve decentralized control, such as decentralized automation and distributed data analytics processes [10, 11]. Furthermore, blockchain technologies are ideal for implementing supply chain management use cases, given the decentralized nature of industrial value chains and the fact that they cannot always operate based on trusted third parties [12]. Also, blockchains are very appealing when it comes to implementing data provenance functionalities, which are among the main pillars of data reliability in industrial environments. In this context, the present chapter provides the following contributions:

- It reviews the use of blockchain technology in manufacturing, through providing a taxonomy of the most prominent manufacturing-oriented use cases. In doing so, our review complements other recent reviews (e.g., [9, 13, 14]).

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- It provides a detailed presentation of blockchain systems for data provenance and traceability in industrial environments. As already outlined, data traceability functionalities are key to ensuring data reliability.
- It introduces a novel blockchain-based approach for AI algorithms and analytics traceability in manufacturing environments. The presented approach goes beyond the traceability of industrial data entities to the provenance of AI algorithms meta, as well as of their outcomes (e.g., analytics and classification outcomes of machine learning techniques). As such it is well suited for supporting trusted AI use cases in manufacturing, which is the overall theme of the book and of subsequent chapters.

To provide the above-listed contributions, the remaining of the chapter is structured as follows: Section 1.2 following this introduction provides an overview of the main use cases in manufacturing. The section ends-up illustrating why data provenance is one of the most important use cases for manufacturing deployments. Section 1.3 reviews the main data provenance and traceability solutions that have been proposed and/or implemented in the research literature. It focused on manufacturing applications, yet some examples from other industrial sectors are also given. Section 1.4 introduces the AI algorithms metadata and AI analytics traceability system. It presents the structure of the blockchain that is used for the implementation of the system, along with the main data and metadata tracked. Section 1.5 is the final section of the chapter. It draws main conclusions and illustrates the connection of the present chapter to other parts of the book.

1.2 Blockchain Applications in Manufacturing

1.2.1 Overview

Blockchain technology is one of the digital enablers of smart manufacturing. It is suitable for applications that require reliable, transparent and secure traceability of data, including: (i) Traceability of industrial data as part of data provenance use cases; (ii) Tracking distributed manufacturing resources towards streamlining production processes that involve multiple actors in the manufacturing chain; (iii) Secure data sharing in the scope of processes that involve exchange of digital models such as Additive Manufacturing (AM) use cases; (iv) Tracking and tracing industrial assets in support of processes like maintenance, repairs, and lifecycle assessment; (v) Ensuring transparency and auditability of manufacturing chains, including coordination of supply chain management and logistic processes; (vi) Increasing the cybersecurity of digital manufacturing infrastructures through eliminating single points of failure and enabling faster updates of IoT devices (e.g., firmware updates, patching); (vii) Development of smart diagnostics and self-service applications for

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Table 1.1. Value propositions of blockchain use cases in manufacturing.

Use Cases Type	Value Propositions
Decentralized Automation	• Reduced Latency for edge computing operations close to the field (automation, analytics) • Trusted data sharing between cloud providers, manufacturers and devices
Secure Information Sharing	• Auditability and transparency of information. • Information consistency based on consensus mechanisms. • Accelerated access to information from the best peer.
Additive Manufacturing	• Secure storage of digital assets. • Trusted and transparent information sharing for IP protection.
Equipment Authentication	• Signing and sealing the interactions of equipment with data. • Ensuring transparency and auditability in the use of assets.
Cybersecurity	• No single points of failure – more difficult to hack. • Automated update of patches and firmware updates through smart contracts.
Conformity to SLAs, Standards and Regulation	• Automated lifecycle stages: discovery and negotiation, deployment, monitoring, billing/penalty, and termination. • Augmented clarity by univocally defining the rules and by recording interactions between physical and non-physical parties in a definitive manner. • Trust in environments where parties do not need to cultivate and maintain relationships of trust among them.

machines, where the machines themselves will be able to monitor their state, diagnose problems, and autonomously place service, consumables replenishment, or part replacement requests to the machine maintenance vendors; (viii) Monitoring conformity to Service Level Agreements (SLAs), Standards and Regulation by translating their mandates into self-imposed smart contracts; (ix) Enabling equipment management, and more specifically identity authentication and authorization; and (x) Providing a framework to largely automate the subprocesses that compose a maintenance equipment leasing procedure, such as two-part negotiation, payment accomplishment and insurance agreement. An overview of relevant use cases and their benefits are presented in Table 1.1.

Following paragraphs present the main blockchain use cases in manufacturing environments.

1.2.2 Decentralized Manufacturing Automation: Intelligence Beyond Cloud-based Manufacturing

Cloud computing is nowadays considered an integral element of most Industry 4.0 deployments. Many manufacturing use cases integrate and analyse data on the

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cloud to enable applications like asset management and digital twins. This cloud-based approach to digital automation has significant advantages stemming from the scalability, capacity and quality of service offered by cloud infrastructures. Nevertheless, it comes also with disadvantages such as: (i) The need to transmit data over a wide area network that results in high latency and is not appropriate for low latency use cases that involve automation operations close to the field; and (ii) the requirement for continuous internet connectivity, which cannot be taken for granted in industrial environments.

Blockchain technology can alleviate the challenges of cloud-based manufacturing through enabling decentralized automation approaches that reduce latency and boost smartness. As a prominent example, in [15] and [10] the authors leverage blockchain technology to implement an edge computing automation paradigm, including distributed automated and distributed data analytics functionalities. As another example, in [16] a blockchain has been used to speed up the flow of production operations based on the coordination of information flows across manufacturing plans and warehouses by means of CPS systems and IoT devices. It falls in the scope of a broader class of systems that aim at deviating from cloud-based manufacturing towards supporting real-time interactions with CPPS systems. Interconnectivity between devices is considered important in this direction [17].

When compared to cloud-based automation systems, blockchains provide also increased trust between interacting actors, including users, CPPS systems and shopfloor services. Specifically, they facilitate trusted data sharing on the shop floor level. This is outlined in [18], which proposes two complementary blockchain-based infrastructures for data sharing: (i) A public blockchain network that is destined to facilitate trusted interactions between cloud providers and manufacturers and (ii) A private blockchain network that boosts trusted data sharing at the shop level, leveraging machine-level connections for data collection.

1.2.3 Secure Information Sharing

Blockchain infrastructures can be used to establish a trusted decentralized environment for sharing data in a secure way. This fosters the implementation of secure information sharing use cases. For instance, a blockchain based system that boosts information sharing for Injection Mould Redesign (IMR) is described in [19]. The system emphasizes trusted information sharing between blockchain participants, while at the same time optimizing the efficiency of the sharing processes through selecting the most appropriate (i.e., faster) peer for accessing the shared knowledge.

In another trusted information sharing use case, a blockchain infrastructure enables the implementation and execution of smart contracts for the sharing of critical information [20]. The blockchain network comprises peers deployed in

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manufacturing machines, system-on-chip platforms, and computing nodes. These peers enable a consortium of disparate organizations to communicate through a decentralized network. Trust is boosted by the application of data provenance mechanisms based on a proper audit trail.

1.2.4 Additive Manufacturing

The implementation of blockchain infrastructure for secure and trusted data exchange is particularly useful for AM applications [21]. The latter leverage digital models of the products and are constantly gaining momentum in the scope of the Industry 4.0 revolution. This is because they improve product design, boost shorter time-to-market, and increase manufacturing agility. One of the biggest challenges of AM is the secure exchange of data across the stakeholders involved in the production process, also given the fact that some of the exchanged data comprise Intellectual Property (IP) (e.g., digital models of a product) and other valuable assets. Blockchain technology facilitates the secure storage and exchange of digital assets, which is the reason why there many blockchain use cases in AM.

In a recent research paper [22], blockchain technology has been used for IP rights management in the context of an AM network. It has been also exploited for monitoring printed parts through their lifecycle, while tracking process improvements. The solution is studied in a broader supply chain context, where the benefits of blockchain (i.e., security data sharing, enhanced visibility, and auditability) are highlighted and acknowledged. Another blockchain application for AM is presented in [23], which emphasizes in the metal additive manufacturing process for components of the aircraft industry. The application is essentially a digital twin that supports the secure end-to-end tracing of data generated during AM processes.

1.2.5 Equipment Identity Management

Distributed ledger technologies are well-suited to provide an effective mechanism that enables equipment management, and more specifically identity authentication and authorization. Quality control requires strict supervision over which equipment has clearance to modify which subsets of data collections. What is more, the logs assembled during the procedure ought to be immutable. By assigning a digital identifier to each piece of equipment (e.g., a sensor in an IoT infrastructure), allowing it to univocally "sign" its interactions with data, transparency and, therefore, an uncontested single source of truth are formulated step by step [24, 25].

In practice, various blockchains and other DLTs provide the possibility of creating unique accounts, fitted with a pair of cryptographic keys; a public one to be

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universally authenticated and a private one to "sign" transactions. In such a configuration, any interaction with data is signed with the equipment's private key and can be verified by anyone who has access to the latter's public key. This verification proves that the equipment had access to the private key, and therefore is likely to be the one associated with the public key. This also ensures that the digital signature has not been tampered with, as it is mathematically bound to the key it originally was made with. From their part, smart contracts can be employed for both handling authorization requests and translating authorization policies into machine-readable self-executing code.

Overall, the use of DLT for equipment identify management, provides the following benefits:

- Security risks related to password authentication are mitigated. For example, there is no possibility for the third party to use a simple/frequent password or to share it unintentionally.
- Authentication of mobile devices, such as phones, tablets, and Augmented Reality (AR) glasses, is less prone to risk. No cookies or other retrievable objects remain on the device.
- Storage and logs are immutable per DLT specifications.
- Weak authentication protocols and human negligence do not pose a threat.
- Security regulations are more severe and privately manageable as opposed to cloud repositories.
- There is no centralized data honeypot for hackers to target.
- There is no need for action if the external user for some reason needs to be un-certified in the future.

1.2.6 Cybersecurity

Many of the previous listed use cases come with security-related value propositions. The latter emerge indirectly e.g., as part of securing data sharing processes. However, blockchain technologies can be used for strengthening the cybersecurity of digital manufacturing infrastructures [14]. For instance, blockchain technology boosts application decentralization, which eliminates single points of failure and boosts the distributional of computational loads across various servers. Likewise, distributed ledger technologies enable decentralized ways for automating the process of updating or patching IIoT devices based on smart contracts [26]. They are also used to provide decentralized trust and accountability without relying on trusted parties [27]. Furthermore, using blockchain technologies data from IIoT devices can be anonymized and remain private within edge nodes i.e., hardly accessible to non-authorized users.

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1.2.7 Monitoring Conformity to Service Level Agreements (SLAs), Standards and Regulation

The advantages of translating Service Level Agreements to self-imposed smart contracts are noteworthy [28]. First and foremost, this process automates their lifecycle stages: discovery and negotiation, deployment, monitoring, billing/penalty and termination. Furthermore, it introduces clarity, since all rules are univocally defined, and transparency, since all interactions between physical and non-physical parties are recorded in a definitive manner. Lastly, DLT technologies are suitable for environments where parties do not need to cultivate and maintain relationships of trust among them [29].

In a real-world application, a "master" smart contract can be designed to enforce legal standards and agreements of any kind. By cross-examining the data uploaded by different stakeholders all parties can verify to what extent the process meets the predefined regulatory conditions. Once all the requirements are met, the regulatory approval may be automatically granted through a smart contract with no further need for on-site inspections or in-person verification.

1.3 Blockchain-based Data Provenance and Traceability

1.3.1 Overview

Data provenance and traceability is one of the most prominent blockchain use cases in industry. It is commonly proposed in cases where several of the following issues hold:

- **Resilience Concerns:** Centralized data provenance databases are more susceptible to hacking and sometimes do not withstand failures. This motivates the use of more decentralized infrastructures.
- **Multiple Writers in different Trust Domains:** There are multiple writers in the data provenance and traceability database, which may raise trust issues. This is particularly relevant in cases where writers belong in different administrative domains and trust domains.
- **Lack of clear rules to Control Data input:** Blockchain infrastructures are suitable for implementing data input rules (including validation rules) by means of consensus mechanisms and smart contracts [30].

1.3.2 Data Provenance Systems

One of the first and most prominent blockchain-based data provenance infrastructures is Provchain [31]. It enables auditing of data operations over cloud storage

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in real-time, while supporting access control and intrusion detection. The infrastructure leverages the tamper-proof properties of blockchain technology by maintaining a decentralized time-stamp log of user operations along with a blockchain receipt. Moreover, it supports privacy preservation features by preventing a direct correlation between users and provenance records, by means of a hashed user ID. Finally, Provchain validates provenance data entries by confirming every block using a blockchain receipt.

There are also solutions that provide traceability and provenance at the level of entire data entities. For instance, the ProductChain blockchain [32] aims at keeping track of processes in the food supply chain. In this direction, a permissioned blockchain is employed along with a transaction vocabulary for the target domain. The system provides interfaces that enable consumers and other stakeholders to access food product provenance information, without disclosing information about trade flows. ProductChain provides very good performance (i.e., query response times) and is therefore suitable for a broader class of supply chain management applications. Beyond product data provenance and traceability in the food chain, there are also blockchain systems for manufacturing chains (e.g., composite materials traceability) [33] and other agricultural products [34]. The latter are recorded in terms of their identity, species name, planting-time, company-name, greenhouse number, and geographical location. Likewise, provenance records about agricultural processes include information about identity, date & time, person, digital-signature, location, operation type, and company.

As another example, the SmartProvenance system leverages smart contracts and consensus mechanism [35] to provide reliable data provenance assuming that the majority of blockchain participants operate properly. The system supports privacy preservation using public key encryption and digital signatures. Blockchain data provenance infrastructures are commonly combined with cloud infrastructures to provide traceability of metadata such as industrial processes configuration information [36].

1.3.3 Gaps for Trusted AI

The above-listed provenance systems provide a sound basis for the implementation of data traceability platforms for reliable BigData analytics in industrial applications. Nevertheless, they are mostly focused on tracing data entities like products and assets without provisions for the provenance of machine learning and deep learning algorithms. The latter is important for the implementation of consistent, reliable, and trusted AI analytics operations in manufacturing environments. AI/ML provenance can help detecting and mitigating cyber-attacks against AI systems such as poisoning. It can also boost the implementation of

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configurable cyber-defence strategies that are grounded on auditing the trustworthiness of AI/ML algorithms training data. The traceability of AI algorithms by means of blockchain infrastructures and smart contracts has a dual flavour:

- **Provenance of AI algorithms metadata and configuration:** Blockchain infrastructures can be used to boost the integrity of AI algorithms and models configurations such as the weights of the neurons of a deep neural network or the parameters of linear regression algorithms. In a trusted AI environment, reliable algorithms that have not been hacked must be used.
- **Provenance of AI analytics outcomes:** Many manufacturing decisions are based on the outcomes of analytics operations. To this end, malicious parties may attempt to compromise the integrity and correctness of analytics outcomes. Blockchain infrastructures can boost the correctness and consistency of analytics outcomes to ensure that manufacturing operations leverage the actual outcomes of trusted AI algorithms i.e., that the outcomes are trusted as well.

1.4 Blockchain Data Provenance for Trusted AI in Manufacturing

1.4.1 Overview and Scope

The EU funded STAR project[1] researches, develops and validates technologies that boost trusted AI in production lines. The project studies technologies that cover many different AI systems in the manufacturing sector, including machine learning and deep learning algorithms, as well as human robot collaboration. It also deals with the safety of these systems such as the safe movement of autonomous mobile robots during their operation in a production plant. The scientific and the technological development areas of the STAR project are presented in other chapters of the book.

Data reliability is an integral element of trusted AI systems in manufacturing, as they are needed for training and operating trusted AI systems. To ensure industrial data reliability, STAR develops a blockchain-based data provenance infrastructure. The latter is destined to leverage the benefits of distributed ledger technologies that were presented in the previous section. The STAR blockchain is not intended to substitute conventional battle-hardened databases. This is because blockchain infrastructures are not best suited for managing large volumes of data

1. H2020 STAR (Safe and Trusted Human Centric Artificial Intelligence in Future Manufacturing Lines) Project, Grant Agreement Number 956573, https://star-ai.eu/

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and transactions. Customization and frequent changes are among blockchain's top enemies. Furthermore, blockchain cannot compete with conventional databases and datastores in performance (e.g., in terms of latency) and responsiveness. Finally, blockchain infrastructures are not suitable for use cases that require reversibility (e.g., rollback operations). Considering the above-listed limitations, the STAR blockchain is implemented as a data infrastructure that complements other data management infrastructures (e.g., databases, datastore) in persisting and manage provenance data (i.e., "meta-data") about data entities like AI algorithms and their analytics outcomes.

Overall, the STAR data management infrastructures exploit the best of both worlds:

(i) State of the art BigData management infrastructures are used to persist large volumes of raw transaction data and to support data operations over them;
(ii) A blockchain infrastructure is used to persist metadata about industrial data towards offering data provenance and traceability functionalities for industrial data entities, including AI algorithms and their outcomes. These metadata boost the cyber-security of the systems and enable the implementation of security risks mitigation strategies.

1.4.2 Performance Considerations and Blockchain Selection

Even though the STAR blockchain is not destined to store and manage large volumes of raw data transactions, it must feature a decent performance for the provenance tasks at hand. Specifically, a blockchain for industrial metadata provenance should provide support for several hundreds of transactions per second, in order to persist the metadata and the outcomes of the AI models that are deployed and executed in a factory.

Public blockchain infrastructures are usually criticized about their poor performance. For instance, the bitcoin protocol is one of the slowest blockchain protocols as new blocks of transactions are validated every ten (10) minutes on average. Ethereum is much faster (i.e., approx. 15 seconds per block are required), yet far from providing decent performance for industrial use cases. This slow performance of public blockchains is due to their complex mining algorithms that safeguard their high security and provide the means for generating new cryptocurrencies. Likewise, this slow performance comes with poor energy efficiency, as mining algorithms are energy intensive.

While public blockchains are not suitable for industrial applications, there is another class of blockchain networks that provides performance suitable for manufacturing use cases. Specifically, this is the case with private permissioned blockchain networks where anyone can participate in the distributed ledger infrastructures as

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soon as it has proper permission from the governing entity of the blockchain. In this blockchain type, the governing entity of the blockchain defines the operations that each one of the participants can perform on the blockchain in terms of creation and execution of transactions and smart contracts. Permissioned blockchain networks consist of a controlled and more limited number of participants. As such they need not operate based on complex mining protocols and Proof-of-Work (PoW) but can rather dispose with lightweight consensus protocols such as Proof-of-Stake (PoS) [Bashir18]. Permissioned blockchains remain slower than ordinary databases and do not scale to 100s of thousands of transactions per second, yet they can achieve performance of few thousands of transactions per seconds. Considering the requirements of the STAR data provenance system, we opted for a private permissioned blockchain. Moreover, we selected the HyperLedger Fabric infrastructure for the implementation of the STAR blockchain [37].

1.4.3 Architecture of Data Provance and Cybersecurity Sub-System

A part of the STAR architecture for secure and trusted AI systems is depicted in Figure 1.1. Specifically, the figure illustrates the sub-system that deals with the reliability of industrial data and the security of AI algorithms in industrial environments. The sub-system sits between: (i) The digital manufacturing platforms and the CPPS systems of an Industry 4.0 shopfloor and (ii) The security teams of the factory, such as Factory IT and cybersecurity experts, as well as CERTs (Computer Emergency Response Teams) and CSIRTs (Computer Security Incident Response Teams). The architecture specifies the following modules:

- **Runtime Monitoring System (RMS) – Data sources connectors and probes:** The RMS provides the means for collecting information from the CPPS systems and the digital manufacturing platforms of the shopfloor. It comprises several configurable data sources connectors and probes, which perform the data acquisition. Connectors and probes are also in charge of capturing the metadata associated with each data source and data capture (e.g., the identifier, the type, and the data formats of the source).
- **Data provenance and traceability applications:** These are decentralized applications (i.e., smart contracts) that run over a permissioned blockchain. They write metadata (i.e., industrial data and AI algorithms configurations) in the various peers of the blockchain. Moreover, they access data configurations that are written in the blockchain, including information about data volumes, statistical data properties, data locations etc. of various data sources. Such decentralized applications are used by the cyber-defence strategies of the sub-systems to identify possible security risks associated with AI algorithms.

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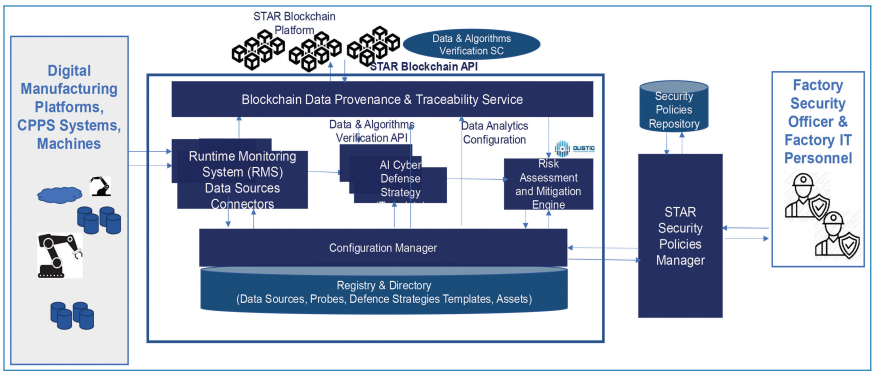


Figure 1.1. Part of the architecture of the STAR system for secure and trusted AI in manufacturing.

- **AI Cyber-defence strategies:** These are security mechanisms that detect security risk and attacks against AI systems, such as poisoning and evasion attacks. They are structured in the form of templates that can be contextualized to different manufacturing environments. In detecting the various attacks, they leverage information about data configurations that are persisted in the blockchain. To this end, they access smart contract functionalities through an appropriate façade and a related API (Application Programming Interface). For instance, a cyber-defence strategy may use information about the statistical properties of the industrial datasets that are used to train an algorithm, to identify a potential poisoning attack. In this case, the blockchain will provide the actual "sealed" statistical properties, which will differ from the poisoned training data.
- **Risk assessment and mitigation engine:** This module accesses the importance of detected risks and proposes actions for their mitigation. It comprises a Security Knowledge Base (SKB) i.e., a repository of known vulnerabilities and attacks. The engine consults the SKB towards providing fast revolution of known attack patterns, as well fast identification of related mitigation actions.
- **Registry of probes, algorithms, templates, and other assets:** To support the configurable and dynamic operation of the system, the various components are registered in a proper catalogue (i.e., registry). The registry provides real-time information about the probes, algorithms and templates that are available, along with information about their status (e.g., active, inactive). Part of the registry is used by the RMS, as presented in following paragraphs of this section.
- **Security policies manager:** This module configures the operation of the subsystem through activating and configurating specific data sources, probes, cyber-defence strategies and AI models. These configurations are provided

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in the form of a security policy, which is activated and enforced based on interactions with the above listed modules. Security experts and teams are responsible for specifying and deploying proper security policies.

1.4.4 Blockchain Network Implementation and Deployment

Figure 1.2 illustrates how the Blockchain Data Provenance and Traceability service interacts with other non-Blockchain modules of the STAR platform.

It exhibits a rather complex architecture, the assemblage of which requires the use of several interconnected machines each hosting some of its components, thus formulating a private permissioned Blockchain network. An Organization participating in the network in this context is a non-Blockchain module of the STAR architecture, such as the RMS or the Configuration Manager, that gains benefit from recording information on the Blockchain. Everything that interacts with the Blockchain network acquires their organizational identity from their digital certificate and their Membership Service Provider (MSP) definition.

Communication of service owners with the Blockchain Network, takes place indirectly, but via a multi-level Backend application that exposes several APIs to client applications. Another choice would be to transfer those functionalities directly to the platform's various service components, which conforms more

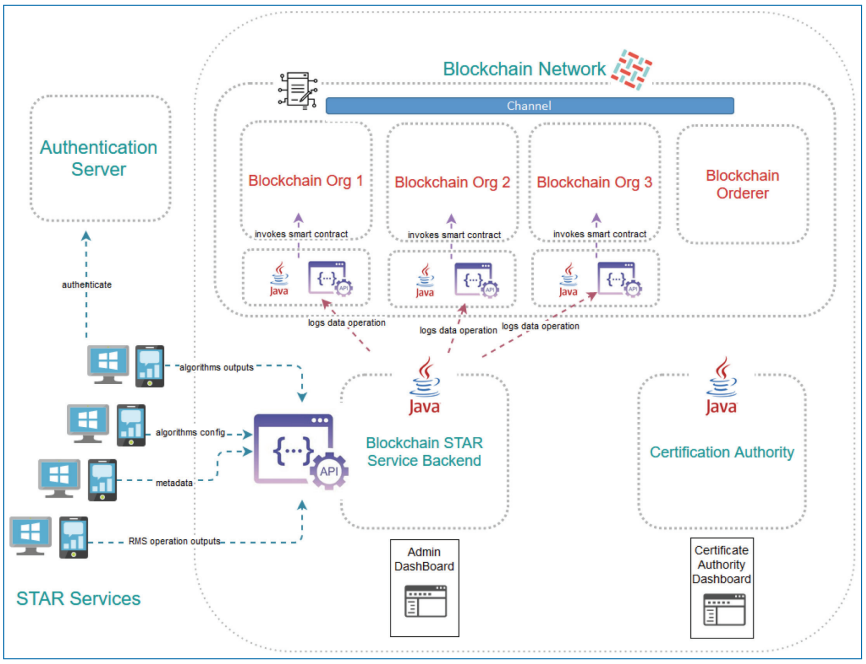


Figure 1.2. STAR data provenance and traceability service.

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naturally with the Blockchain decentralization paradigm, but this would have required their developers to have extensive expertise in decentralized applications development. A final proposal would be to make Smart Contracts handle everything a back-end process does, including the job of the APIs. However, assigning only specific tasks to each distinct component has been judged to be way cleaner and manageable.

The various APIs serve different users: one exists for the Authority tasked with producing the certificates that will allow participation on the network. Another serves administrative and monitoring tasks. The most important API serves the parties recording and retrieving data. To conclude, users are authenticated against an identity management server (for instance Keycloak[2]), which entitles them to access the permissioned Blockchain network.

Figure 1.3 provides an anatomy of the Permissioned Blockchain network implementation of the project, shedding light into the various technologies used to materialize its components. The building blocks and operational processes of the Blockchain are directly dictated by the Hyperledger Fabric architectural paradigm. Specifically, the architecture exhibits a two-levels structure: (i) A first level that comprises different administrative entities (i.e., modules synthesizing the STAR platform) and (ii) A second level that comprises various peer nodes (i.e., sub-components of said modules) within each service. In-line with the Fabric's architecture, the various peers can interact and exchange data through one or more Channels. Only the peers that participate in a Channel can communicate through it and share joint ownership of the information stored on the Blockchain. This provides flexibility in clustering the peers in different groups that engage in various groups of disjoint transactions. Every node can participate in several Channels i.e., it can communicate with different groups of peer nodes. One or more Smart Contracts, which in the STAR context are describing traceability information and algorithm configurations, are deployed across a Channel.

Each (peer) node maintains a ledger of the transactions where it is involved. To this end, each peer maintains a database such as Apache CouchDB.[3] Every time the global state commonly maintained via the Blockchain changes (e.g., new metadata about a data source become available) a new transaction is initiated, through an interaction with a Smart Contract. The latter is responsible for consistently changing the status of said state to reflect the inclusion of the new information. In the Fabric infrastructure, some nodes can propose and endorse transactions, while others are only able to propose them. Nodes that can endorse transactions ought to

2. Keycloak Open Source Identity and Access Management System, https://www.keycloak.org/

3. Apache CouchDB, https://couchdb.apache.org/

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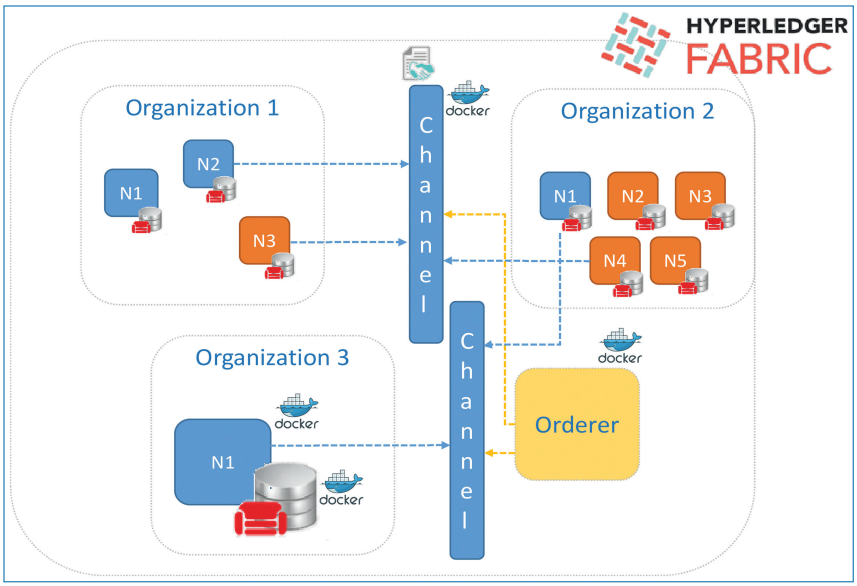


Figure 1.3. STAR permissioned blockchain network example.

reach an agreement on the current global state of the data by employing a consensus algorithm i.e., Raft[4] in our case. This increases the flexibility of the permissions and functionalities that can be granted to the different nodes, which represent a data source within digital manufacturing platforms and CPPS systems.

As specified in the Fabric paradigm, special nodes (called "Orderers") validate the various requests to update the state against the existing configuration, generate new configuration transactions, and package them into blocks that are relayed to all peers on the Channel. The peers then process the configuration transactions in order to verify that the modifications approved by the Orderers do indeed satisfy the policies defined in the Channel.

1.4.5 Data Modelling and Persistence

To implement industrial data provenance, there is a need for using a proper data model of the industrial metadata that will be stored in the Blockchain. In this direction, STAR extends background digital models of the authors [38], which comprise the following main metadata:

- **Data Source Definition (DSD):** Defines the properties of a data source on the shop floor, such as a data stream from a sensor, a CPPS, or an automation device.

4. Raft Consensus Algorithm, https://raft.github.io/

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- **Data Interface Specification (DI):** The DI is associated with a data source and provides the information need to connect to it and access its data, including details like network protocol, port, the network address and more.
- **Data Kind (DK):** Specifies the semantics of the data source. The DK can be used to define virtually any type of data in an open and extensible way.
- **Data Source Manifest (DSM):** Specifies a specific instance of a data source in-line with its DSD, DI and DK specifications. Multiple manifests (i.e. DSMs) are therefore used to represent the data sources that are available in the factory.
- **Observation:** Models the actual dataset that stems from an instance of a data source that is represented through a DSM. Hence, it references a DSM, which drives the specification of the types of the attributes of the Observation in-line with the DK that facilitates the discoverability of the data. An Observation is associated with a timestamp and keeps track of the location of the data source in case it is associated with mobile device or CPPS (e.g., mobile robot). The value type of observation is a complex object which is described with the DK entity that an Observation references. Hence, an observation can depict multiple raw measurements coming from a machine or a single value (i.e., the number of cycles/m of a rotor) or even an Analytics result (i.e., the calculated Remaining Useful Life (RUL) of a machine).
- **Edge Gateway:** Models an edge gateway of an edge computing deployment i.e., a deployment following the edge computing paradigm. In the scope of an edge computing deployment, data sources are associated with an edge gateway. This usually implies not only a logical association but a physical association as well, i.e. an edge gateway is deployed at a station and manages data sources in close physical proximity to the station.

The above entities are used to represent the data sources of a digital shopfloor in a modular, dynamic, and extensible way. This is based on the registry of data sources and their manifests, which keeps track of the various data sources that register to it. Furthermore, to facilitate the management and configuration of analytics functions and workflows over the various data sources, several analytics-related entities are also specified, including:

- **Analytics Processor Definition (APD):** Specifies a processing function to be applied on one or more data sources. Three processing functions are defined, including functions that pre-process the data of a data source (i.e. Pre-Processors), functions that store the outcomes of the processing (i.e. Store Processors) and functions that analyse the data from the data sources (i.e. Analytics Processors). These three types of processors can be combined in

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various configurations over the data sources in order to define different analytics workflows.

- **Analytics Processor Manifest (APM):** Represents an instance of a processor that is defined through the APD. The instance specifies the type of processors and its actual logic through linking to a programming function (e.g., Java).
- **Analytics Orchestrator Manifest (AOM):** Represents an entire analytics workflow. It defines a combination of analytics processor instances (i.e. of APMs) that implements a distributed data analytics task. The latter can span multiple edge gateways and operate over their data sources.

The digital models for industrial data provenance and data analytics follow a hierarchical structure, which defines the different relationships between the various entities. For example, an edge gateway comprises multiple data source manifests. Each one of the latter is associated with a data source definition. Likewise, Observations are associated with instances of data sources i.e. data sources manifests [Kefalakis19], [Soldatos19].

Also, the digital models presented above offer some special characteristics in order to be adaptable in various AI-based application specific scenarios such as predictive maintenance, quality management and zero defect manufacturing (ZDM). The Entities that facilitate such extensions are the Data Kind (DK), Observations and Additional Information.

Data Kind specifies the semantics of the data source data, which provides flexibility in modelling different types of data. It can be used to define virtually any type of data in an open and extensible way. It describes the type, format and data kind of the values that are produced by an AI system. Specifically, the "kind" of the data is represented with the QuantityKind attribute which is an abstract classifier that represents the concept of "kind of quantity". A QuantityKind represents the essence of a quantity without any numerical value or unit. (e.g. A sensor -sensor1- measures temperature: sensor1 has quantityKind temperature). The Data Kind is not only used to describe data sources that are used by an AI system, but also data sources that are produced from the system i.e., the outcomes of AI/ML analytics.

Observation entities model the actual data that stem from an instance of a data source (i.e., modelled through a DSM). Hence, it references a DSM, which drives the specification of the types of the attributes of the Observation in-line with the DK. An Observation is associated with a timestamp and keeps track of the location of the data source in case it is associated with a mobile (rather than a stationary) data source. Hence, it has a location attribute as well. Observation holds the measurement or result of a Data Source at the "value" entity which is of type anyType. This means that it can support any type of value (even complex structures) that are identified from the Data Kind it is referencing. Similar to Data Kind and Data

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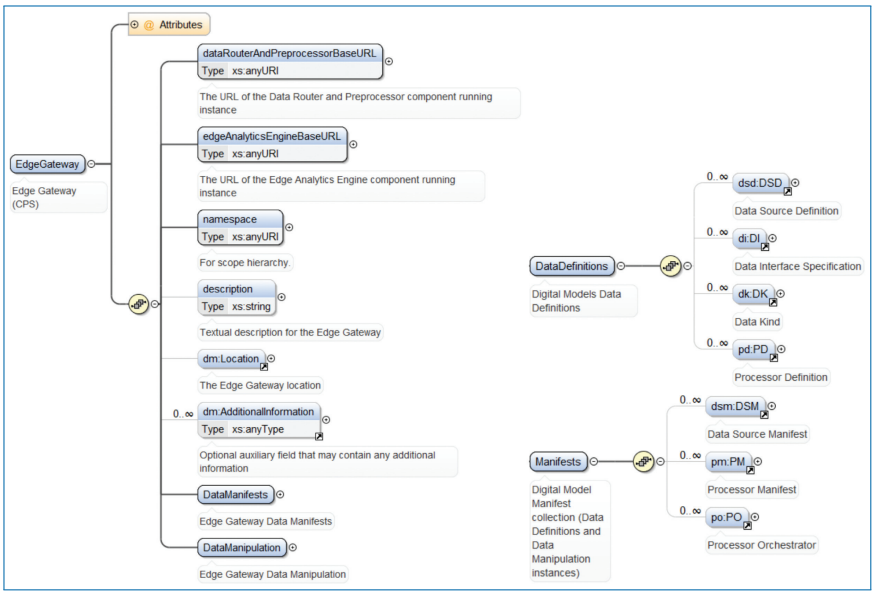


Figure 1.4. Snapshot of the digital models metadata.

Source the Observation is not only used to describe data that are captured by an AI system but also data that are produced from the system (i.e., AI analytics produce Observations).

Finally, AdditionalInformation is a generic entity which allows the extension of the existing data model with additional attributed that may be required. For instance, AdditionalInformation entities are used in the EdgeGateway and Core digital model structures to provide optional auxiliary fields that can be used for further extensions.

Figure 1.4 presents a snapshot of the different entities that are used by the different components. These are persisted in the blockchain to support provenance and traceability functionalities ate different levels and for different AI applications. For example, the Edge Gateway data entities provide discoverability of the Edge systems and enable configuration and management of the data provenance system. DSM entities provide a global and local visibility of the different industrial data sources (i.e. DSDs) used by the AI systems. This enables also discoverability of DSDs and of the Observed data that they comprise. Furthermore, the DSD can be used at configuration time to associate the data source with some AI algorithm and its outcomes. Finally, the persistence of AOM information facilitates the configuration of AI analytics functions and enables the dynamic discovery of analytics outcomes (i.e., Observations) based on the AOM they are associated with i.e., based on the AI data and algorithms that produced these observations.

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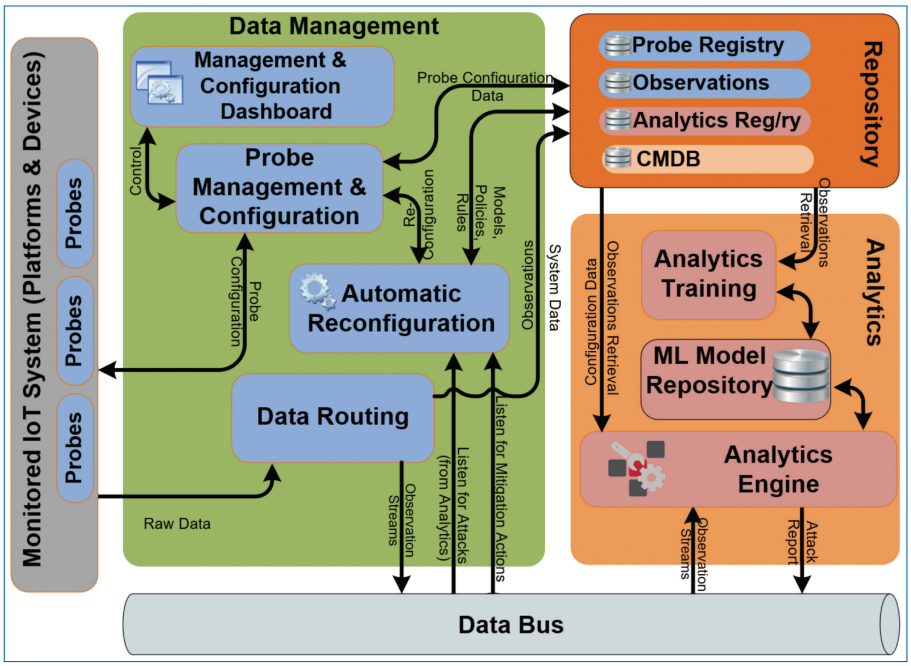


Figure 1.5. Run-time monitoring system in the trusted AI infrastructure.

1.4.6 Run-Time Monitoring System

The Runtime Monitoring System (RMS) collects industrial data in real-time and stores them for further processing and analysis by AI algorithms. RMS features lightweight monitoring probes that are responsible for the data collection and publishing to AI platforms. The RMS provides configuration and management mechanisms over the monitoring probes as well as data models and data transformation engines that will enable the discoverability and reusability of the collected data. The probe management is facilitated by an internal probe registry that maintains information about the probes (including their status), while enabling probe creation, reconfiguration, and discovery. Figure 1.5 illustrates a functional diagram of the main system components.

The main functionalities and interactions of these components are as follows:

- **Data bus:** This is a communications channel that routes real time data. Platform components may subscribe to the data bus to receive data of specific interest.
- **Deployed probes:** Probes collect data from the target IoT system or application and stream them to the IoT platform through the data routing component.

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- **Probe Management and Configuration:** This module is responsible for managing and configuring the deployed probes. It can receive automatic probe configuration commands and correspondingly configures the managed probes. Manual probe configuration commands may also be received through the dashboard. The Management and Configuration dashboard provides a user interface to the Probe Management and Configuration component.
- **Probe Registry:** Maintains a record of the deployed probes. Probe deployment data, as well as state and configuration data are maintained by the registry. The registry provides probe creation, reconfiguration, and search capabilities. It facilitates the automatic deployment of probes and their dynamic discovery.
- **Automatic Reconfiguration:** This sends automatic probe re-configuration commands in-line with the implemented security policy.
- **Data Storage:** This contains historic security data that have been collected by the deployed probes. These data can be used by the Data Analytics to train itself and produce a set of security templates that will be used subsequently for identifying security issues on the target IoT system.
- **Configuration Management Database (CMDB):** This is part of the data storage. It contains information about all assets of the RMS, including their attributes and configuration parameters.

RMS is implemented based on technologies of the Elastic Stack, as outlined in Figure 1.6.

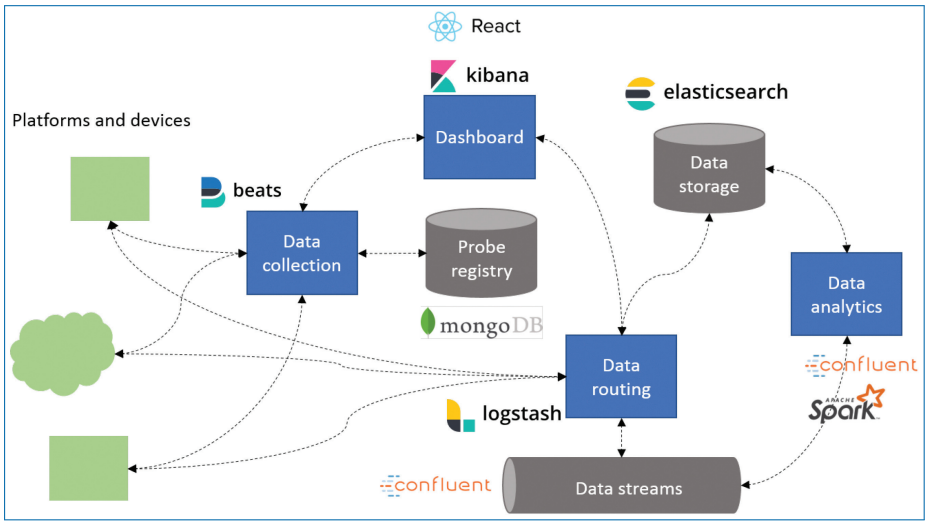


Figure 1.6. Implementation technologies of the run-time monitoring system.

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1.5 Conclusion

Blockchain technologies offers advantages for data provenance and traceability in industrial environments. These advantages stem from the data security, tampered-proof and decentralized nature of distributed ledger technologies. In recent years, the research community has developed and demonstrated various blockchain-based systems for secure information sharing and data provenance in the scope of manufacturing applications. These systems provide the means for tracking industrial data entities (e.g., products, assets), as well as processes performed over them. Nevertheless, they lack support for tracking and tracing AI algorithms, models, and their analytics outcomes. This is a set-back for their use in the emerging wave of trusted AI applications. As part of this chapter, we have presented a novel blockchain infrastructure that supports data provenance for AI models and algorithms towards boosting trusted AI in industrial environments. The presented blockchain provides a foundation for data reliability. As such it blends nicely with AI systems that are presented in later chapters of the book.

The presented blockchain infrastructure is in its early implementation stages. In addition to completing its implementation and validation, we plan to benchmark its performance and scalability in real-life manufacturing environments. Moreover, we will collect feedback from manufacturing stakeholders, including practitioners with field experience. This will lead us to conclusions about the practical applicability of blockchain technology in production lines. Likewise, it will help the research community identify the next steps that could move blockchain technology from the realm of pilot experiments to practical enterprise scale deployments in industrial environments.

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